

Discrete Structures

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Why Study Discrete Math?

Discrete Mathematics is the study of discrete (not continuous) objects

▷ Unlike Calculus (no limits, continuity, differentiation)



source: Google images

Why Study Discrete Math?

Computer Science = Problem Solving

- Mathematics is at the heart of problem solving
 - definition, modeling, solution, analysis
- Computers are discrete objects operating one discontinuous (discrete) step at a time
- Discrete mathematics is the right framework for describing precisely CS concepts

👤 : I don't like **math**,
I'm just going to college
to choose "**IT**" major.

💻 "**IT**" Major : Welcome to
Calculus, Linear Algebra,
Mathematical logic, Statistics,
Numerical Methods, Discrete
mathematics.

source: ProgrammerHumor.io

Computer Science is no more about computers than astronomy is about telescopes.

Edsger Dijkstra

Why Study Discrete Math?

Conceptual core of computer science is **Algorithm**

Discrete Mathematics helps provide

- techniques to design algorithms
- methods to prove the correctness of algorithms
- tools to analyze algorithms

Recommended Books

Text Book

- Discrete Mathematics and Its Applications

Kenneth H. Rosen

Including Solution Manual

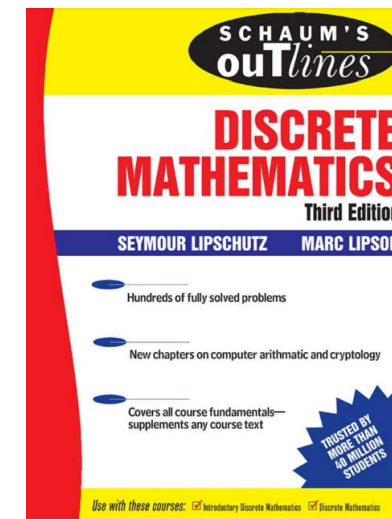
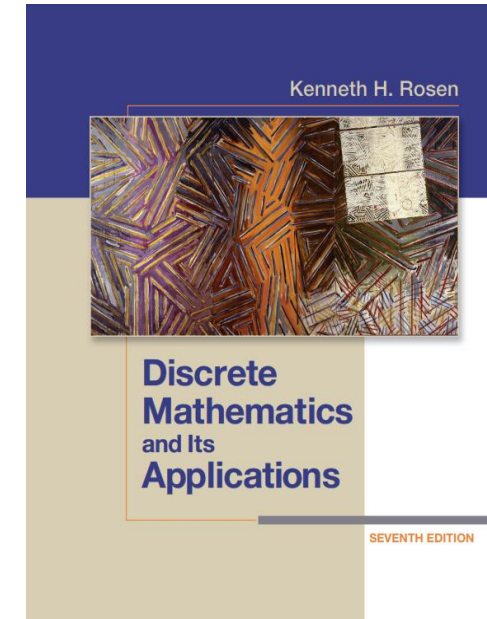
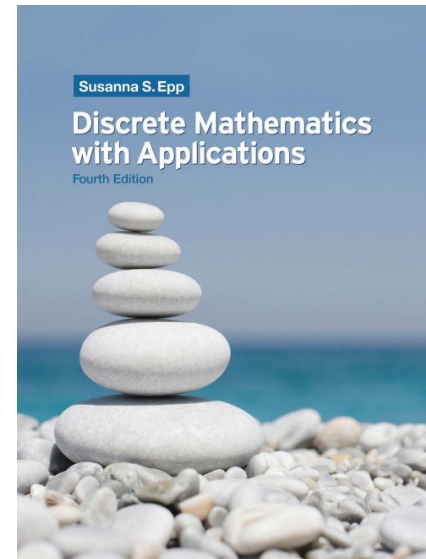
Reference Book:

- Discrete Mathematics with Applications

Susanna S.Epp

- Discrete Mathematics

Schaums' Outlines



Why Study Discrete Math? Topics



Logic:

- Fundamental to all mathematical disciplines
- Useful for hardware design, computer architecture
- Artificial Intelligence, Natural Language Processing, Knowledge Representation

Logic: Logic is the study of reasoning. It is the basic on which all the sentence are build. OR Correct and Incorrect reasoning.

A **proposition** is a declarative sentence (that is, a sentence that declares a fact) that is either true or false, but not both.

Imperative

Declarative

Interrogative

1.What time is it?

2.Read this carefully.

3. $x + 1 = 2$.

4. $x + y = z$

5.I am in Class.

6.Cat is bigger than Rat.

7. $1 + 2 = 5$

8. $2 + 2 = 5$

Propositional Logic:

The area of logic that deals with propositions.

- “Elephants are bigger than mice.”

Is this a statement? **yes**

Is this a proposition? **yes**

What is the truth value
of the proposition? **true**

- “ $y > 5$ ”

Is this a statement? **yes**

Is this a proposition? **no**

Truth value?

Depends on the value of y , but this
value is not specified.

Compound Propositions

Logical Connectives

I like A.I

Logical operators are used to form new propositions also called compound propositions from two or more existing propositions.

- Negation $\neg$
- Conjunction $\wedge$
- Disjunction $\vee$
- Implication $\rightarrow$
- Biconditional $\leftrightarrow$

- Conjunction (AND)
- Disjunction (OR)
- Exclusive or (XOR)
- Implication (if – then)
- Biconditional (if and only if)

A **truth table** is a tabular representation of all the combinations of values for inputs and their corresponding outputs

Negation

- I am in NOT class
- Cat cannot fly.
- Today is Sunday.

The Truth Table for the Negation of a Proposition.	
p	$\neg p$
T	F
F	T

Conjunction

Let p and q be propositions. The conjunction of p and q , denoted by $p \wedge q$, is the proposition “ p and q ”. The conjunction $p \wedge q$ is true when both p and q are true and is false otherwise.

Today is Friday AND it is raining today.
I'd like pizza AND a salad for lunch.

The Truth Table for the Conjunction of Two Propositions.		
p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction

Let p and q be propositions. The disjunction of p and q , denoted by $p \vee q$, is the proposition “ p or q ”. The disjunction $p \vee q$ is false when both p and q are false and is true otherwise.

Inclusive or : The disjunction is true when at least one of the two propositions is true.

“Students who have taken calculus or computer science can take this class.”

Exclusive or : The disjunction is true only when one of the proposition is true.

“Ice cream or pudding will be served after lunch.”

The Truth Table for the Disjunction of Two Propositions.		
p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

The Truth Table for the Exclusive Or (XOR) of Two Propositions.		
p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

CONDITIONAL STATEMENT / IMPLICATION:

Let p and q be propositions. The conditional statement $p \rightarrow q$, is the proposition “if p , then q ” which is false when p is true and q is false, and true otherwise.

p is called the hypothesis (or antecedent or premise) and q is called the conclusion (or consequence).

Example: “If I am elected, then I will lower taxes.”

elected, lower taxes.

not elected, lower taxes.

not elected, not lower taxes.

elected, not lower taxes.

TABLE 5 The Truth Table for the Conditional Statement $p \rightarrow q$.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

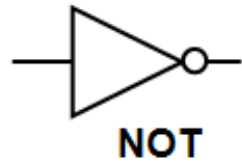
CONDITIONAL STATEMENT / Bi-implications:

Let p and q be propositions. The biconditional statement $p \leftrightarrow q$ is the proposition “ p if and only if q .” The biconditional statement $p \leftrightarrow q$ is true when p and q have the same truth values, and is false otherwise. Biconditional statements are also called bi-implications

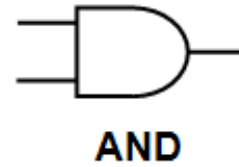
“You can take the flight if and only if you buy a ticket.”

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

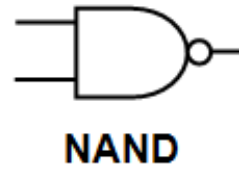
1.2.6 Logic Circuits – Logic Gate



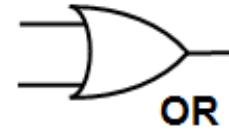
Input	Output
I	F
0	1
1	0



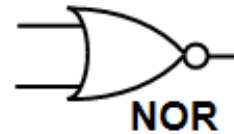
Inputs		Output
A	B	F
0	0	0
1	0	0
0	1	0
1	1	1



Inputs		Output
A	B	F
0	0	1
1	0	1
0	1	1
1	1	0



Inputs		Output
A	B	F
0	0	0
1	0	1
0	1	1
1	1	1



Inputs		Output
A	B	F
0	0	1
1	0	0
0	1	0
1	1	0

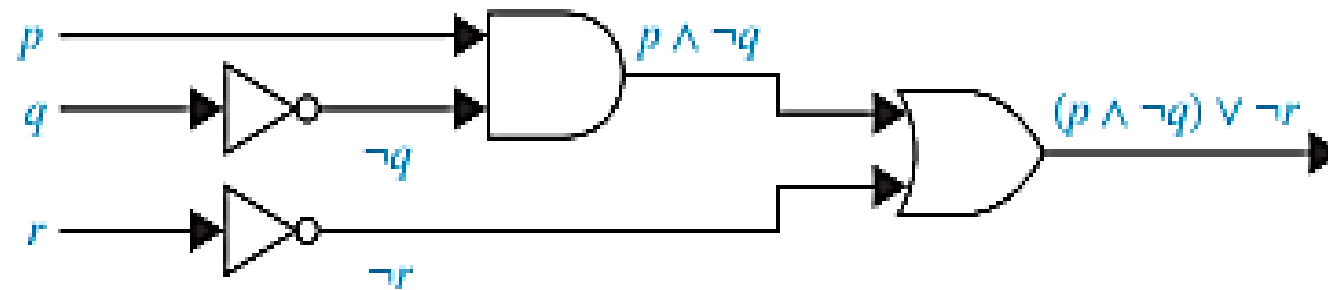


Inputs		Output
A	B	F
0	0	0
0	1	1
1	0	1
1	1	0

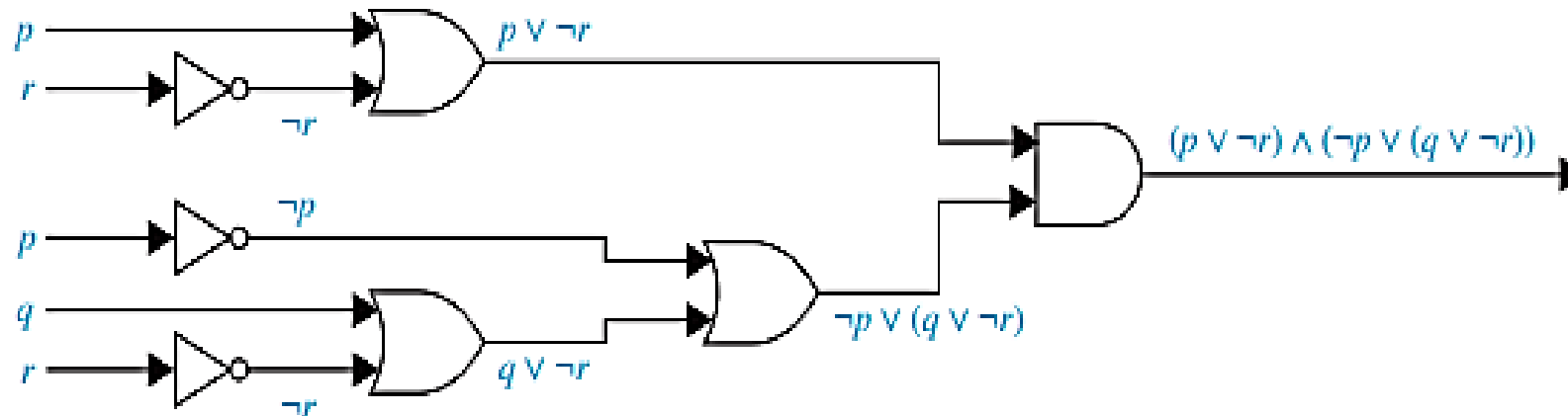


Inputs		Output
A	B	F
0	0	1
0	1	0
1	0	0
1	1	1

1.2.6 Logic Circuits – Logic Gate



$$(p \wedge \neg q) \vee \neg r$$



$$(p \vee \neg r) \wedge (\neg p \vee (q \vee \neg r)).$$

Other Conditional Statements:

Conditional: “if p , then q ” (or $p \rightarrow q$)

- If you are working hard, then you are a topper.

Converse: “if q , then p ” (or $q \rightarrow p$)

- If you are a topper, then you are working hard.

Inverse: “if not p , then not q ” (or $\neg p \rightarrow \neg q$)

- If you are not working hard, then you are not a topper.

Contrapositive: “if not q , then not p ” (or $\neg q \rightarrow \neg p$)

- If you are not topper, then you are not working hard.

Show using a truth table that the conditional is equivalent to the contrapositive.

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg q \rightarrow \neg p$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	$\textcircled{\text{T}}$	T

Show using truth tables that neither the **converse** nor **inverse** of an implication are not equivalent to the implication.

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg p \rightarrow \neg q$	$q \rightarrow p$
T	T	F	F	T	T	T
T	F	F	T	F	T	T
F	T	T	F	T	F	F
F	F	T	T	$\textcircled{\text{T}}$	T	T

CONDITIONAL STATEMENT:

Different Ways of Expressing $p \rightarrow q$

Let p and q be propositions. The conditional statement $p \rightarrow q$ is the proposition “if p , then q .” The conditional statement $p \rightarrow q$ is false when p is true and q is false, and true otherwise. In the conditional statement $p \rightarrow q$, p is called the hypothesis (or antecedent or premise) and q is called the conclusion (or consequence).

“if p , then q ”

“if p , q ”

“ p is sufficient for q ”

“ q if p ”

“ q when p ”

“a necessary condition for p is q ”

“ q unless $\neg p$ ”

“ p implies q ”

“ p only if q ”

“a sufficient condition for q is p ”

“ q whenever p ”

“ q is necessary for p ”

“ q follows from p ”

Example: “If I am elected, then I will lower taxes.”

elected, lower taxes.

not elected, lower taxes.

not elected, not lower taxes.

elected, not lower taxes.

“p only if q”

This is often the most confusing one because “if” is right next to q. However, in logic, “only if” introduces the consequence (the result), not the condition.

The Logic: It means p cannot happen unless q is true. If you have p, you are guaranteed to have q.

Example: “You can drive only if you have a license.”

(a) If you are driving (p), then you must have a license (q).

(b) It does not mean if you have a license, you are currently driving.

“q unless not p”

The word “unless” can be translated as “if not”. So, “q unless not p” can be written as: “q if not (not p).”

The Logic: Two negatives cancel each other out ($\neg(\neg p) \equiv p$).

The Transformation:

1. Start with: q unless $\neg p$
2. Replace “unless” with “if not”: q if $\neg(\neg p)$
3. Simplify: q if p
4. Rearrange: If p, then q

Example: “The alarm will ring (q) unless the power is not on ($\neg p$).”

This means: If the power is on (p), the alarm will ring (q).

Practice Time

Write each of these statements in the form “if p, then q” in English.

[Hint: Refer to the list of common ways to express conditional statements provided in this section.]

- a) You send me an e-mail message only if I will remember to send you the address.
- b) To be a citizen of this country, it is sufficient that you were born in the Pakistan.
- c) If you keep your textbook, it will be a useful reference in your future courses.
- d) The Red Wings will win the Stanley Cup if their goalie plays well.
- e) That you get the job implies that you had the best credentials.
- f) The beach erodes whenever there is a storm.
- g) It is necessary to have a valid password to log on to the server.
- h) You will reach the summit unless you begin your climb too late.

Practice Time

Let p , q , and r be the propositions

p : You have the flu.

q : You miss the final examination.

r : You pass the course.

Express each of these propositions as an English sentence.

a) $p \rightarrow q$

b) $\neg q \leftrightarrow r$

c) $q \rightarrow \neg r$

d) $p \vee q \vee r$

e) $(p \rightarrow \neg r) \vee (q \rightarrow \neg r)$

f) $(p \wedge q) \vee (\neg q \wedge r)$

Practice Time

Let p , q , and r be the propositions

p : You get an A on the final exam.

q : You do every exercise in this book.

r : You get an A in this class.

Write these propositions using p , q , and r and logical connectives.

- a) You get an A in this class, but you do not do every exercise in this book.
- b) You get an A on the final, you do every exercise in this book, and you get an A in this class.
- c) To get an A in this class, it is necessary for you to get an A on the final.
- d) You get an A on the final, but you don't do every exercise in this book; nevertheless, you get an A in this class.
- e) Getting an A on the final and doing every exercise in this book is sufficient for getting an A in this class.
- f) You will get an A in this class if and only if you either do every exercise in this book or you get an A on the final.

Propositional Equivalences, Law of Logic

- A **tautology** is a proposition which is always true.

Example: $p \vee \neg p$

- A **contradiction** is a proposition which is always false.

Example: $p \wedge \neg p$

- A **contingency** is a proposition which is neither a tautology nor a contradiction, such as p

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

- Two compound propositions p and q are logically equivalent if $p \leftrightarrow q$ is a tautology.
- We write this as $p \Leftrightarrow q$ or as $p \equiv q$ where p and q are compound propositions.
- Two compound propositions p and q are equivalent if and only if the columns in a truth table giving their truth values agree.
- This truth table show $\neg p \vee q$ is equivalent to $p \rightarrow q$.

p	q	$\neg p$	$\neg p \vee q$	$p \rightarrow q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

Propositional Equivalences, Law of Logic

Show that the proposition form $p \vee \neg p$ is a tautology and the proposition form $p \wedge \neg p$ is a contradiction.

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

Exercise: If t is a tautology and c is contradiction, show that $p \vee t \equiv t$ and $p \wedge c \equiv c$?

Contradiction occurs when we get a statement p , such that p is true and its negation $\neg p$ is also true.

Truth Tables For Compound Propositions

Operator	Precedence
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

TABLE 7 The Truth Table of $(p \vee \neg q) \rightarrow (p \wedge q)$.					
p	q	$\neg q$	$p \vee \neg q$	$p \wedge q$	$(p \vee \neg q) \rightarrow (p \wedge q)$
T	T	F	T	T	T
T	F	T	T	F	F
F	T	F	F	F	T
F	F	T	T	F	F

Construct a truth table for $p \vee q \rightarrow \neg r$

p	q	r	$\neg r$	$p \vee q$	$p \vee q \rightarrow \neg r$
T	T	T	F	T	F
T	T	F	T	T	T
T	F	T	F	T	F
T	F	F	T	T	T
F	T	T	F	T	F
F	T	F	T	T	T
F	F	T	F	F	T
F	F	F	T	F	T

Propositional Equivalences, Law of Logic

TABLE 6 Logical Equivalences.

<i>Equivalence</i>	<i>Name</i>
$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Identity laws
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

De Morgan's laws state that:

The negation of an **and** proposition is logically equivalent to the **or** proposition in which each component is negated.

$$1. \neg(p \wedge q) \equiv \neg p \vee \neg q$$

The negation of an **or** proposition is logically equivalent to the **and** proposition in which each component is negated.

$$2. \neg(p \vee q) \equiv \neg p \wedge \neg q$$

Question: Negate the following compound Propositions

- John is six feet tall **and** he weights at least 200 pounds.
- The bus was late **or** Tom's watch was slow.

Solution

- John is **not** six feet tall **or** he weighs **less** than 200 pounds.
- The bus was **not** late **and** Tom's watch was **not** slow.

Propositional Equivalences, Law of Logic

TABLE 5 A Demonstration That $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ Are Logically Equivalent.

p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T
F	T	F	F	F	T	F	F
F	F	T	F	F	F	T	F
F	F	F	F	F	F	F	F

TABLE 7 Logical Equivalences Involving Conditional Statements.

$$\begin{aligned}
 p \rightarrow q &\equiv \neg p \vee q \\
 p \rightarrow q &\equiv \neg q \rightarrow \neg p \\
 p \vee q &\equiv \neg p \rightarrow q \\
 p \wedge q &\equiv \neg(p \rightarrow \neg q) \\
 \neg(p \rightarrow q) &\equiv p \wedge \neg q \\
 (p \rightarrow q) \wedge (p \rightarrow r) &\equiv p \rightarrow (q \wedge r) \\
 (p \rightarrow r) \wedge (q \rightarrow r) &\equiv (p \vee q) \rightarrow r \\
 (p \rightarrow q) \vee (p \rightarrow r) &\equiv p \rightarrow (q \vee r) \\
 (p \rightarrow r) \vee (q \rightarrow r) &\equiv (p \wedge q) \rightarrow r
 \end{aligned}$$

TABLE 8 Logical Equivalences Involving Biconditional Statements.

$$\begin{aligned}
 p \leftrightarrow q &\equiv (p \rightarrow q) \wedge (q \rightarrow p) \\
 p \leftrightarrow q &\equiv \neg p \leftrightarrow \neg q \\
 p \leftrightarrow q &\equiv (p \wedge q) \vee (\neg p \wedge \neg q) \\
 \neg(p \leftrightarrow q) &\equiv p \leftrightarrow \neg q
 \end{aligned}$$

Propositional Equivalences, Law of Logic

Using laws of logic, show that

$$\neg(\neg p \wedge q) \wedge (p \vee q) \equiv p.$$

Solution

Take $\neg(\neg p \wedge q) \wedge (p \vee q)$

$$\begin{aligned} &\equiv (\neg(\neg p) \vee \neg q) \wedge (p \vee q) && \text{by Demorgan's Law} \\ &\equiv (p \vee \neg q) \wedge (p \vee q) && \text{by Double Negation Law} \\ &\equiv p \vee (\neg q \wedge q) && \text{by Distributive Law} \\ &\equiv p \vee F && \text{by Negation Law} \\ &\equiv p && \text{by Identity Law} \end{aligned}$$

TABLE 6 Logical Equivalences.

<i>Equivalence</i>	<i>Name</i>
$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Identity laws
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

Propositional Equivalences, Law of Logic

Show $\neg(p \rightarrow q)$ is equivalent to $p \wedge \neg q$.

Show that $\neg(p \vee \neg(p \wedge q))$ is a contradiction

TABLE 6 Logical Equivalences.

<i>Equivalence</i>	<i>Name</i>
$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Identity laws
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

Propositional Equivalences, Law of Logic

$$\sim(\sim p \wedge q) \wedge (p \vee q) \equiv p$$

TABLE 6 Logical Equivalences.

<i>Equivalence</i>	<i>Name</i>
$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Identity laws
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
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$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

Propositional Equivalences, Law of Logic

Prove that $\neg[r \vee (q \wedge (r \rightarrow \neg p))] \equiv \neg r \wedge (p \vee \neg q)$

$$\neg[r \vee (q \wedge (r \rightarrow \neg p))]$$

$$\equiv \neg r \wedge \neg(q \wedge (r \rightarrow \neg p)),$$

$$\equiv \neg r \wedge \neg(q \wedge (r \vee \neg p)),$$

$$\equiv \neg r \wedge \neg(q \wedge (r \vee \neg p)),$$

$$\equiv \neg r \wedge (\neg q \vee \neg(r \vee \neg p)),$$

$$\equiv \neg r \wedge (\neg q \vee (r \wedge p)),$$

$$\equiv (\neg r \wedge \neg q) \vee (\neg r \wedge (r \wedge p)),$$

$$\equiv (\neg r \wedge \neg q) \vee ((\neg r \wedge r) \wedge p),$$

$$\equiv (\neg r \wedge \neg q) \vee (\neg r \wedge p),$$

$$\equiv \neg r \wedge (\neg q \vee p),$$

$$\equiv \neg r \wedge (p \vee \neg q),$$

De Morgan's law

Conditional rewritten as disjunction

Double negation law

De Morgan's law

De Morgan's law, double negation

Distributive law

Associative law

Idempotent law

Distributive law

Commutative law

TABLE 6 Logical Equivalences.

Equivalence	Name
$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Identity laws
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

$$(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$$

p	q	r	$p \vee \neg q$	$q \vee \neg r$	$r \vee \neg p$	$(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$
T	T	T	T	T	T	T
T	T	F	T	T	F	F
T	F	T	T	F	T	F
T	F	F	T	T	F	F
F	T	T	F	T	T	F
F	T	F	F	T	T	F
F	F	T	T	F	T	F
F	F	F	T	T	T	T

Notation

$\bigvee_{j=1}^n p_j$ is used for $p_1 \vee p_2 \vee \dots \vee p_n$

$\bigwedge_{j=1}^n p_j$ is used for $p_1 \wedge p_2 \wedge \dots \wedge p_n$

Predicates

Predicate Logic deals with predicates, which are propositions, consist of variables.

A predicate is a sentence that contains a finite number of variables and becomes a statement when specific values are substituted for the variables. The domain of a predicate variable is the set of all values that may be substituted in place of the variable.

Property of subject expression that contain variables

Why do we study it in DM, [to deal with variables]

Propositional Function

Predicate is also called propositional Function

Denote: $P(\text{Variable})$ or $P(x)$

- **Example 1:** Let $P(x)$ denote the statement " $x > 10$ ". What are the truth values of $P(11)$ and $P(5)$?

Solution: $P(11)$ is equivalent to the statement $11 > 10$, which is True.

$P(5)$ is equivalent to the statement $5 > 10$, which is False.

- **Example 2:** Let $R(x, y)$ denote the statement " $x = y + 1$ ". What is the truth value of the propositions $R(1, 3)$ and $R(2, 1)$?

Solution: $R(1, 3)$ is the statement $1 = 3 + 1$, which is False.

$R(2, 1)$ is the statement $2 = 1 + 1$, which is True.

Predicates

Consider the propositional function $Q(x, y, z)$ defined by:

$$x + y = z$$

What is the truth value of $Q(2, 3, 5)$?

True

What is the truth value of $Q(0, 1, 2)$?

False



$$2+3=5$$

$$5=5$$



$$0+1=2$$

$$1=2$$

Predicates

EXAMPLE 4 Let $A(c, n)$ denote the statement “Computer c is connected to network n ,” where c is a variable representing a computer and n is a variable representing a network. Suppose that the computer MATH1 is connected to network CAMPUS2, but not to network CAMPUS1. What are the values of $A(\text{MATH1}, \text{CAMPUS1})$ and $A(\text{MATH1}, \text{CAMPUS2})$?

Solution: Because MATH1 is not connected to the CAMPUS1 network, we see that $A(\text{MATH1}, \text{CAMPUS1})$ is false. However, because MATH1 is connected to the CAMPUS2 network, we see that $A(\text{MATH1}, \text{CAMPUS2})$ is true. ◀

Similarly, we can let $R(x, y, z)$ denote the statement “ $x + y = z$.” When values are assigned to the variables x , y , and z , this statement has a truth value.

EXAMPLE 5 What are the truth values of the propositions $R(1, 2, 3)$ and $R(0, 0, 1)$?

Solution: The proposition $R(1, 2, 3)$ is obtained by setting $x = 1$, $y = 2$, and $z = 3$ in the statement $R(x, y, z)$. We see that $R(1, 2, 3)$ is the statement “ $1 + 2 = 3$,” which is true. Also note that $R(0, 0, 1)$, which is the statement “ $0 + 0 = 1$,” is false. ◀

In general, a statement involving the n variables $x_1, x_2, \dots, x_n$ can be denoted by

$$P(x_1, x_2, \dots, x_n).$$

A statement of the form $P(x_1, x_2, \dots, x_n)$ is the value of the **propositional function** P at the n -tuple $(x_1, x_2, \dots, x_n)$, and P is also called an **n -place predicate** or a **n -ary predicate**.

Universe of Discourse

We need to have a universe of discourse

UoD: A collection of subjects; things we are talking about

With a clear UoD, for each value of the variable predicates become propositions

Each variable may have a different universe of discourse

Examples of UoD

What is an appropriate UoD for the variable x in

$P(x) : x > 4$?

- a) Set of people
- b) Set of real numbers
- c) Set of cats

What is an appropriate UoD for the variable x in

$P(x) : x \text{ is a professor at LUMS}$?

- a) Set of people
- b) Set of real numbers
- c) Set of cats

Examples of UoD

What are appropriate UoDs for the variables x and y in,

$Q(x, y) : x > y$?

- a) Set of people and set of real numbers
- b) Set of real numbers and set of cats
- c) Set of real numbers and set of real numbers

What are appropriate UoDs for the variables x, y , and z in,

$Q(x, y, z) : x \text{ teaches course } y \text{ in university } z$?

- a) People, Courses, Universities
- b) Courses, Universities, People
- c) University, People, Courses

Predicates and Quantifiers

The variable of predicates is quantified by quantifiers. There are two types of quantifier in predicate logic - **Existential Quantifier** and **Universal Quantifier**.

TABLE 1 Quantifiers.		
Statement	When True?	When False?
$\forall x P(x)$	$P(x)$ is true for every x .	There is an x for which $P(x)$ is false.
$\exists x P(x)$	There is an x for which $P(x)$ is true.	$P(x)$ is false for every x .

- $\exists x P(x)$ is read as “For some x , $P(x)$ ”, or as “There is an x such that $P(x)$,” or “For at least one x , $P(x)$.”

Examples:

1. If $P(x)$ denotes “ $x > 0$ ” and U is the integers, then $\exists x P(x)$ is true. It is also true if U is the positive integers.
2. If $P(x)$ denotes “ $x < 0$ ” and U is the positive integers, then $\exists x P(x)$ is false.
3. If $P(x)$ denotes “ x is even” and U is the integers, then $\exists x P(x)$ is true.

Quantifiers

- **Example 1:** Let $P(x)$ be the statement " $x + 2 > x$ ". What is the truth value of the statement $\forall x P(x)$?

Solution: As $x + 2$ is greater than x for any real number, so $P(x) \equiv T$ for all x or $\forall x P(x) \equiv T$.

-
- **Example :** Let $P(x)$ be the statement " $x > 5$ ". What is the truth value of the statement $\exists x P(x)$?

Solution: $P(x)$ is true for all real numbers greater than 5 and false for all real numbers less than 5. So $\exists x P(x) \equiv T$.

Quantifiers

- $\exists!x P(x)$ means that $P(x)$ is true for one and only one x in the universe of discourse.
- This is commonly expressed in English in the following equivalent ways:
 - “There is a unique x such that $P(x)$.”
 - “There is one and only one x such that $P(x)$ ”
- Examples:
 1. If $P(x)$ denotes “ $x + 1 = 0$ ” and U is the integers, then $\exists!x P(x)$ is true.
 2. But if $P(x)$ denotes “ $x > 0$,” then $\exists!x P(x)$ is false.

Quantifiers

Let $C(x)$ be the statement “ x has a cat,” let $D(x)$ be the statement “ x has a dog,” and let $F(x)$ be the statement “ x has a ferret.” Express each of these statements in terms of $C(x)$, $D(x)$, $F(x)$, quantifiers, and logical connectives. Let the domain consist of all students in your class.

a) A student in your class has a cat, a dog, and a ferret.

Solution: $\exists x (C(x) \wedge D(x) \wedge F(x))$

b) All students in your class have a cat, a dog, and a ferret.

Solution: $\forall x (C(x) \wedge D(x) \wedge F(x))$

c) Some student in your class has a cat and a ferret, but not a dog.

Solution: $\exists x (C(x) \wedge F(x) \wedge \neg D(x))$

d) No student in your class has a cat, a dog, and a ferret.

Solution: $\forall x \neg(C(x) \wedge D(x) \wedge F(x))$

37. Express each of these statements using predicates and quantifiers.

- a) A passenger on an airline qualifies as an elite flyer if the passenger flies more than 25,000 miles in a year or takes more than 25 flights during that year.
- b) A man qualifies for the marathon if his best previous time is less than 3 hours and a woman qualifies for the marathon if her best previous time is less than 3.5 hours.
- c) A student must take at least 60 course hours, or at least 45 course hours and write a master's thesis, and receive a grade no lower than a B in all required courses, to receive a master's degree.
- d) There is a student who has taken more than 21 credit hours in a semester and received all A's.

Answer:

$$a) \forall x (f(x) \vee g(x) \rightarrow e(x))$$

$$b) \forall x (m(x) \wedge c(x, 3.5)) \vee (\neg m(x) \wedge c(x, 3.5)) \rightarrow q(x)$$

$$c) \forall x (M(x) \rightarrow ([H(x, 60) \vee (H(x, 45) \wedge W(x))] \wedge G(x)))$$

$$d) \exists x (C(x) \wedge A(x))$$

Solution

a)

$f(x)$: x is flying more than 25000 miles in a year

$g(x)$: x takes more than 25 flights during that year

$e(x)$: x qualifies as an elite flyer

b)

$m(x)$: x is a man

$c(x, y)$: the best previous time of x is less than y hours

$q(x)$: x qualifies for the marathon

c)

$M(x)$: x receives a Master's degree

$H(x, n)$: x takes at least n course hours

$W(x)$: x writes a Master's thesis

$G(x)$: x receives a grade no lower than B in all courses

d)

$C(x)$: x taken more than 21 credit hours in a semester

$A(x)$: x has received all A

Negating Quantified Expressions

TABLE 2 De Morgan's Laws for Quantifiers.

<i>Negation</i>	<i>Equivalent Statement</i>	<i>When Is Negation True?</i>	<i>When False?</i>
$\neg \exists x P(x)$	$\forall x \neg P(x)$	For every x , $P(x)$ is false.	There is an x for which $P(x)$ is true.
$\neg \forall x P(x)$	$\exists x \neg P(x)$	There is an x for which $P(x)$ is false.	$P(x)$ is true for every x .

Quantifiers, De Morgan's Laws for Quantifiers

What are the negation of the statement “There is an honest politician”?

Solution: Let $H(x)$ denote “ x is honest.” Then the statement “There is an honest politician” is represented by $\exists x H(x)$, where the domain consists of all politicians. The negation of this statement is $\neg \exists x H(x)$, which is equivalent to $\forall x \neg H(x)$.

This negation can be expressed as “Every politician is dishonest.”

(Note: In English, the statement “All politicians are not honest” is ambiguous. In common usage, this statement often means “Not all politicians are honest.”

Consequently, we do not use this statement to express this negation.)

Quantifiers, De Morgan's Laws for Quantifiers

What are the negation of the statement “All Americans eat cheeseburgers”?

Quantifiers, De Morgan's Laws for Quantifiers

What are the negations of the statements $\forall x(x^2 > x)$ and $\exists x(x^2 = x)$?

Solution:

The negation of $\forall x(x^2 > x)$ is the statement $\neg \forall x(x^2 > x)$, which is equivalent to $\exists x \neg(x^2 > x)$. This can be rewritten as $\exists x(x^2 \leq x)$.

The negation of $\exists x(x^2 = x)$ is the statement $\neg \exists x(x^2 = x)$, which is equivalent to $\forall x \neg(x^2 = x)$. This can be rewritten as $\forall x(x^2 \neq x)$.

The truth values of these statements depend on the domain.

Show that $\neg \forall x(P(x) \rightarrow Q(x))$ and $\exists x(P(x) \wedge \neg Q(x))$ are logically equivalent.

Solution:

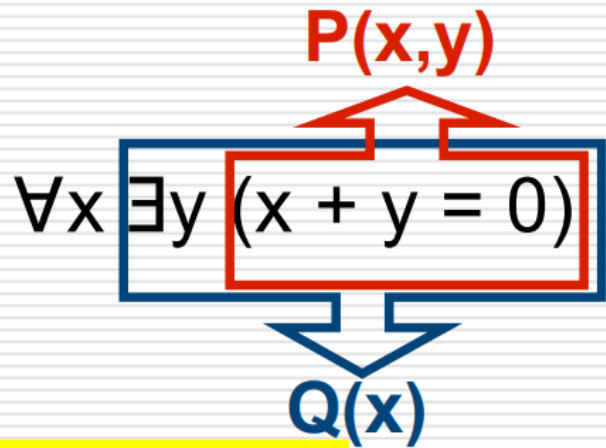
By De Morgan's law for universal quantifiers, we know that $\neg \forall x(P(x) \rightarrow Q(x))$ and $\exists x(\neg(P(x) \rightarrow Q(x)))$ are logically equivalent.

By the fifth logical equivalence, we know that $\neg(P(x) \rightarrow Q(x))$ and $P(x) \wedge \neg Q(x)$ are logically equivalent for every x . Because we can substitute one logically equivalent expression for another in a logical equivalence, it follows that $\neg \forall x(P(x) \rightarrow Q(x))$ and $\exists x(P(x) \wedge \neg Q(x))$ are logically equivalent.

Nested Quantifiers

Two quantifiers are nested if one is within the scope of the other.

$$\forall x \exists y (x + y = 0)$$



$$\forall x Q(x)$$

$$Q(x) \text{ is } \exists y P(x, y)$$

$$P(x, y) \text{ is } (x + y = 0)$$

Examples:

$$\forall x \forall y (x + y = y + x)$$

$$\forall x \forall y \forall z (x + (y + z) = (x + y) + z)$$

$$\forall x \forall y ((x > 0) \wedge (y < 0) \rightarrow (xy < 0))$$

Nested Quantifiers, Orders

TABLE 1 Quantifications of Two Variables.

<i>Statement</i>	<i>When True?</i>	<i>When False?</i>
$\forall x \forall y P(x, y)$ $\forall y \forall x P(x, y)$	$P(x, y)$ is true for every pair x, y .	There is a pair x, y for which $P(x, y)$ is false.
$\forall x \exists y P(x, y)$	For every x there is a y for which $P(x, y)$ is true.	There is an x such that $P(x, y)$ is false for every y .
$\exists x \forall y P(x, y)$	There is an x for which $P(x, y)$ is true for every y .	For every x there is a y for which $P(x, y)$ is false.
$\exists x \exists y P(x, y)$ $\exists y \exists x P(x, y)$	There is a pair x, y for which $P(x, y)$ is true.	$P(x, y)$ is false for every pair x, y .

$\forall x \forall y P(x, y)$

$\forall x \exists y P(x, y)$

$\exists x \forall y P(x, y)$

$\exists x \exists y P(x, y)$

So, we concluded the new concept of Equivalency here as well.

Nested Quantifiers, Questions

Translate these statements into English, where the domain for each variable consists of all real numbers.

a) $\exists x \forall y (xy = y)$

b) $\forall x \forall y ((x > 0) \wedge (y < 0) \rightarrow (xy < 0))$

c) $\forall x \forall y \exists z (x = y + z)$

Solution:

- a) This says that there exists a real number x such that for every real number y , the product xy equals y . That is, there is a multiplicative identity for the real numbers. This is a true statement, since $x = 1$ is the identity.
- b) This statement says that for every real number x and for every real number y , if $x > 0$ and $y < 0$, then $xy < 0$. That is, this statement says that for real numbers x and y , if x is positive and y is negative, then xy is negative. This can be stated more succinctly as “The product of a positive real number and a negative real number is always a negative real number.”
- c) This says that for every pair of real numbers x and y , there exists a real number z that is their sum. In other words, the real numbers are closed under the operation of addition, another true fact. (Some authors would include the uniqueness of z as part of the meaning of the word *closed*.)

Nested Quantifiers, Questions

Negating Nested Quantifiers

1.4.30 Rewrite each of these statements so that negations appear only within predicates (that is, so that no negation is outside a quantifier or an expression involving logical connectives).

- a. $\neg \exists x \exists y p(x, y)$
- b. $\neg \forall x \exists y p(x, y)$
- c. $\neg \exists y (Q(y) \wedge \forall x \neg R(x, y))$
- d. $\neg \exists y (\exists x R(x, y) \vee \forall x S(x, y))$
- e. $\neg \exists y (\forall x \exists z T(x, y, z) \vee \exists x \forall z U(x, y, z))$

a	$\forall x \forall y \neg p(x, y)$
b	$\exists x \forall y \neg p(x, y)$
c	$\forall y (\neg Q(y) \vee \exists x R(x, y))$
d	$\forall y (\forall x \neg R(x, y) \wedge \exists x \neg S(x, y))$
e	$\forall y (\exists x \forall z \neg T(x, y, z) \wedge \forall x \exists z \neg U(x, y, z))$

Order of quantifiers is extremely important

$\forall x \exists y P(x, y)$ **is not the same as** $\exists y \forall x P(x, y)$

$P(x, y) : x$ teaches course y in LUMS

- $\forall x \exists y P(x, y)$: For every instructor, there is a course that (s)he teaches
- $\exists y \forall x P(x, y)$: There is a course that every instructor teaches
- $\exists x \forall y P(x, y)$: There is an instructor who teaches all the courses
- $\forall y \exists x P(x, y)$: For every course, there is an instructor who teaches it

Order of quantifier is extremely important

$$\forall x \exists y P(x, y) \quad \text{is not the same as} \quad \exists y \forall x P(x, y)$$

$Q(x, y, z)$: x is an instructor of course y in university z

- $\forall z \forall y \exists x Q(x, y, z)$: In every uni. for every course there is an instructor
- $\forall z \exists x \forall y Q(x, y, z)$: In every uni. there is an instructor for all courses
- $\exists x \forall z \forall y Q(x, y, z)$: There is an instructor for every course in every uni.

Example

1				
		1	1	
1				
				1
				1

Table: **1**

1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1

Table: **2**

- $\forall \text{ row } x \exists \text{ column } y A[x][y] = 1$
- Every row has at least one 1
- Both tables satisfy this
- $\exists \text{ column } y \forall \text{ row } x A[x][y] = 1$
- There is a column with all 1's
- Table **1** does not satisfy this
- Table **2** satisfy this

Example

1				
		1	1	
1				
				1
				1

Table: 1

1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1

Table: 2

- $\forall \text{ row } x \forall \text{ column } y A[x][y] = 1$
- Every cell has a 1
- Table 1 does not satisfy this
- Table 2 satisfy this

- $\forall \text{ column } y \forall \text{ row } x A[x][y] = 1$
- Every cell has a 1
- Table 1 does not satisfy this
- Table 2 satisfy this

Example

1				
		1	1	
1				
				1
				1

Table: 3

		1		

Table: 4

- $\exists \text{ row } x \exists \text{ column } y A[x][y] = 1$
- There is a 1 in the table
- Both tables satisfy this

- $\exists \text{ column } x \exists \text{ row } y A[x][y] = 1$
- There is a 1 in the table
- Both tables satisfy this

Order of quantifier is extremely important

$\forall x \exists y P(x, y)$ **is not the same as** $\exists y \forall x P(x, y)$



source: <http://wtkrieger.faculty.noctrl.edu/csc230-winter2019/etc/>

If both quantifiers are the same then order does not matter

$\forall x \forall y P(x, y) \equiv \forall y \forall x P(x, y)$ and $\exists x \exists y P(x, y) \equiv \exists y \exists x P(x, y)$

Example

Express each of these system specifications using predicates with UoD's, quantifiers, and logical connectives

ICP 2-30

There are at least two paths connecting every two distinct endpoints on the network

$C(p, x, y)$: Path p connects endpoint x and y

$$\forall x \forall y \neq x \exists p_1 \exists p_2 \neq p_1 C(p_1, x, y) \wedge C(p_2, x, y)$$

Logic Puzzle

Smullyan posed many puzzles about an island that has two kinds of inhabitants, knights, who always tell the truth, and their opposites, knaves, who always lie.

You encounter two people A and B. What are A and B if A says “B is a knight” and B says “The two of us are opposite types?”

Logic Puzzle

Solution: Let p and q be the statements that A is a knight and B is a knight, respectively, so that $\neg p$ and $\neg q$ are the statements that A is a knave and B is a knave, respectively.

We first consider the possibility that A is a knight; this is the statement that p is true. If A is a knight, then he is telling the truth when he says that B is a knight, so that q is true, and A and B are the same type. However, if B is a knight, then B 's statement that A and B are of opposite types, the statement $(p \wedge \neg q) \vee (\neg p \wedge q)$, would have to be true, which it is not, because A and B are both knights. Consequently, we can conclude that A is not a knight, that is, that p is false.

If A is a knave, then because everything a knave says is false, A 's statement that B is a knight, that is, that q is true, is a lie. This means that q is false and B is also a knave. Furthermore, if B is a knave, then B 's statement that A and B are opposite types is a lie, which is consistent with both A and B being knaves. We can conclude that both A and B are knaves. 